

Figure 1. One platform, two cooperating modes, in a live energy OT environment

SOC Mode = deception-based active defence. Research Mode = offline analyst copilot. They share one data layer: live attacker behaviour informs research, research insight informs defence.

Energy operator OT environment (Purdue model)

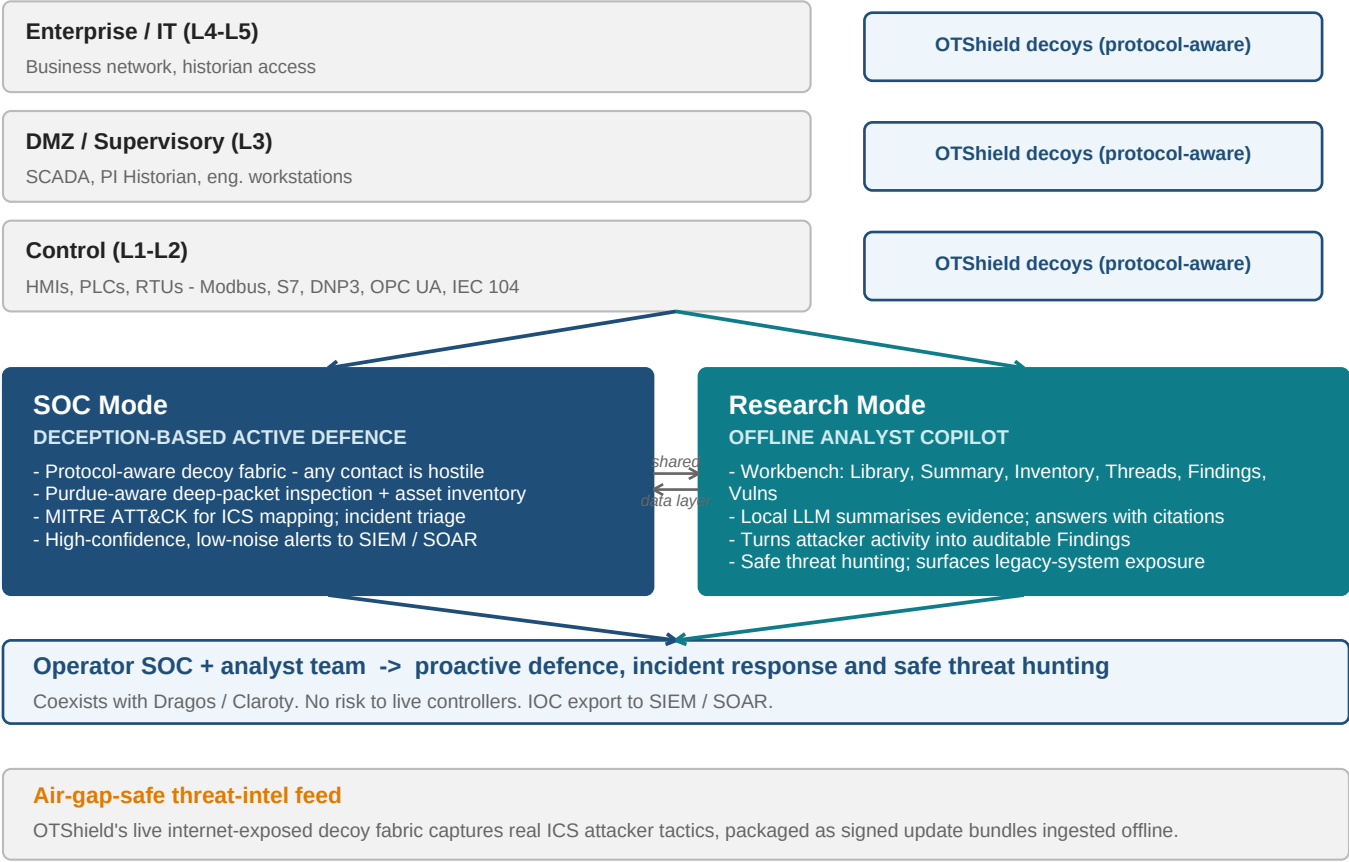
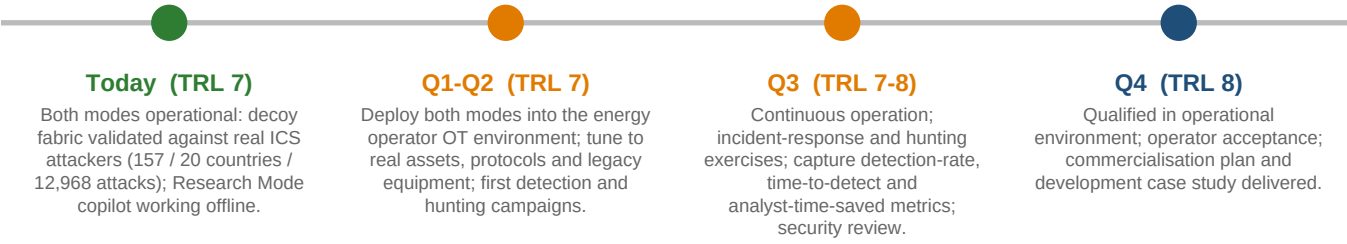


Table 1. Mapping to the four competition challenges

| Challenge                                       | Mode          | How OTShield delivers it in this project                                                                                                  |
|-------------------------------------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Proactive CNI defence & incident response    | SOC Mode      | High-confidence decoy detections + DPI; alerts to SIEM/SOAR; triage and response with the operator SOC team.                              |
| 2. Detection & eviction of sophisticated actors | SOC Mode      | Surfaces protocol-aware intent (unauthorised writes, interlock overrides, remote discovery) mapped across ATT&CK for ICS.                 |
| 3. Threat hunting for CNI organisations         | Research Mode | Copilot lets analysts query detections, manuals and asset data together and follow leads safely with zero risk to live controllers.       |
| 4. Legacy system vulnerabilities in CNI         | Both modes    | SOC Mode inventories and risk-scores legacy assets; Research Mode copilot reasons over them to show how they could be reached and abused. |

Figure 2. Readiness: TRL 7 today -> TRL 8 at project end



**Why it meets the challenge**

One integrated platform - SOC Mode for deception-based active defence, Research Mode for the analyst copilot - is proven, accepted and qualified inside a real UK energy operator, delivering all four challenge areas and ready to scale across the sector.